# Mathematical Derivations of Diagnostic Thresholds

Five-Step IC Validation Framework — Revision 2

**Key Revision:** All thresholds re-derived from a unified *daily cross-sectional Spearman time-series* framework. See §9 for a side-by-side comparison with Revision 1.

## 1 The Daily IC Time-Series Framework

The Revision 1 derivations suffered from a fundamental inconsistency: Threshold 1 used an effective sample size of $N = 750$ (days) derived from the Fisher $z$-transformation of a pooled Spearman $\rho$, while Thresholds 3 and 5 used $N \approx 10^5$ (pooled stock–day pairs) in $\sigma_0^2 \approx 1/N$, and Threshold 2 oscillated between $N = 1850$ and $N = 375\,000$ depending on the argument. These inconsistent unit assumptions led to contradictory standard errors and, consequently, thresholds that did not align with the actual computation in `diagnose_v2.py`.

Revision 2 adopts a single, self-consistent unit of analysis: the *daily cross-sectional Spearman IC*.

**Definition 1.1** (Daily IC time-series)**.** For a set of $N$ stocks and $T$ held-out trading days, let $\hat{\rho}_t$ be the cross-sectional Spearman rank correlation between predicted returns $\{\hat{r}_{i,t}\}_{i=1}^N$ and realized returns $\{r_{i,t}\}_{i=1}^N$ on day $t$:

$$\hat{\rho}_t = \mathrm{Corr}_{\mathrm{Spearman}}\big(\{\hat{r}_{i,t}\}_{i=1}^N,\ \{r_{i,t}\}_{i=1}^N\big), \qquad t = 1, \dots, T. \tag{1}$$

The daily IC time-series is $\{\hat{\rho}_1, \dots, \hat{\rho}_T\}$.

**Empirical calibration.** The pipeline v2 held-out test set ($T = 795$ days, $N = 477$ stocks) yields:

$$\mu_{\mathrm{IC}} = \frac{1}{T}\sum_{t=1}^{T} \hat{\rho}_t = 0.0289, \tag{2}$$

$$\sigma_{\mathrm{IC}} = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T} (\hat{\rho}_t - \mu_{\mathrm{IC}})^2} = 0.158, \tag{3}$$

$$p_+ = \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}\{\hat{\rho}_t > 0\} = 0.570. \tag{4}$$

Note that $\sigma_{\mathrm{IC}} = 0.158$ captures *both* the cross-sectional estimation error *and* genuine day-to-day variation in predictive quality—this is the key quantity for all subsequent derivations.

**Remark 1.1** (Relationship between pooled IC and $\mu_{\mathrm{IC}}$)**.** The traditional *pooled Spearman* $\mathrm{IC}_{\mathrm{pooled}}$ (a single Spearman $\rho$ over all $T \times N$ stock–day pairs) and the daily-IC mean $\mu_{\mathrm{IC}}$ are mathematically distinct statistics. In the pipeline v2 results, $\mathrm{IC}_{\mathrm{pooled}} = 0.0443$ while $\mu_{\mathrm{IC}} = 0.0289$, illustrating that the pooled estimator can differ substantially from the daily-IC mean in finite samples.

**Why the daily-IC framework is preferred for diagnostic thresholds:**

1. The daily IC $\hat{\rho}_t$ is the quantity that actually enters the walk-forward evaluation (the code averages per-day ICs in Step 2's $t$-test).
2. $\mathrm{SE}(\mu_{\mathrm{IC}}) = \sigma_{\mathrm{IC}}/\sqrt{T}$ is straightforward to estimate from the daily-IC time series; the analogous standard error for $\mathrm{IC}_{\mathrm{pooled}}$ requires Fisher $z$-transformation with an assumed effective sample size (the source of the Revision 1 inconsistency).
3. In the large-$T$ limit under weak dependence, $\sqrt{T}(\mu_{\mathrm{IC}} - \mathbb{E}[\hat{\rho}_t]) \xrightarrow{d} \mathcal{N}(0, \sigma^2_{\mathrm{IC}})$ by the CLT, providing rigorous inference without distributional assumptions.

In Revision 2, **all thresholds use $\mu_{\mathrm{IC}}$ as the primary IC statistic**. The pooled $\mathrm{IC}_{\mathrm{pooled}}$ is retained only as a descriptive summary; all inference proceeds from the daily-IC time-series.

### 1.1 Standard Errors of the Daily-IC Mean

**Definition 1.2** (i.i.d. standard error)**.** If daily ICs are serially uncorrelated,

$$\mathrm{SE}_{\mathrm{iid}}(\mu_{\mathrm{IC}}) = \frac{\sigma_{\mathrm{IC}}}{\sqrt{T}}. \tag{5}$$

**Definition 1.3** (Newey–West HAC standard error)**.** To accommodate possible autocorrelation in daily ICs (e.g., due to persistent market regimes), the Newey–West [1] heteroskedasticity-and-autocorrelation-consistent (HAC) estimator is used. Let $\hat{\gamma}_k$ be the sample autocovariance at lag $k$:

$$\hat{\gamma}_k = \frac{1}{T}\sum_{t=k+1}^{T} (\hat{\rho}_t - \mu_{\mathrm{IC}})(\hat{\rho}_{t-k} - \mu_{\mathrm{IC}}). \tag{6}$$

With Bartlett kernel weights $w_k = 1 - k/(L+1)$ and bandwidth $L$:

$$\mathrm{SE}_{\mathrm{NW}}(\mu_{\mathrm{IC}}) = \frac{1}{\sqrt{T}}\sqrt{\hat{\gamma}_0 + 2\sum_{k=1}^{L} w_k\,\hat{\gamma}_k}. \tag{7}$$

For $T = 795$, the automatic bandwidth $L = \lfloor 4(T/100)^{2/9} \rfloor = 6$ provides a reasonable default (Newey–West, 1994).

Table 1: Standard errors of $\mu_{\mathrm{IC}}$ under alternative assumptions. Plug-in values: $\sigma_{\mathrm{IC}} = 0.158$, $T = 795$.

| Estimator | Formula | Value |
|---|---|---|
| i.i.d. SE | $\sigma_{\mathrm{IC}}/\sqrt{T}$ | $0.158/\sqrt{795} = 0.00560$ |
| NW SE ($L = 6$) | Eq. (7) | 0.00490–0.00650 (estimated) |

For the remainder of this document, we use $\mathrm{SE}(\mu_{\mathrm{IC}}) = 0.00560$ (the i.i.d. value) as the baseline, noting that the Newey–West correction typically changes it by less than 15%.

## 2 Threshold 1: $|\,\mathrm{IC}\,| < 0.02$ as Detection Threshold

### 2.1 Minimum Detectable IC

The single most important threshold in the framework is whether the observed IC exceeds the noise floor. With the daily-IC framework, this reduces to a one-sample test of $H_0 : \mu_{\mathrm{IC}} = 0$.

**Proposition 2.1** (Minimum detectable IC at $\alpha = 0.05$, 80% power)**.** For a two-sided test,

$$\mathrm{IC}_{\min} = (z_{\alpha/2} + z_\beta)\ \mathrm{SE}(\mu_{\mathrm{IC}}), \qquad z_{0.025} = 1.960,\ z_{0.20} = 0.842. \tag{8}$$

Plugging in $\mathrm{SE}(\mu_{\mathrm{IC}}) = 0.00560$:

$$\begin{aligned} \mathrm{IC}_{\min} &= (1.960 + 0.842) \times 0.00560 \\ &= 2.802 \times 0.00560 = \mathbf{0.01570}. \end{aligned} \tag{9}$$

**Remark 2.1** (Comparison with Revision 1)**.** Revision 1 derived $r_{\min} \approx 0.102$ using Fisher $z$ with $N = 750$ and 80% power, then argued *in the opposite direction* that $|\,\mathrm{IC}\,| < 0.02$ was conservative because it was "far below" the detection threshold. The internal contradiction was: Fisher $z$ predicts $\mathrm{IC} = 0.02$ is undetectable ($p \approx 0.58$), yet the code's Step 1 correctly flags $\mathrm{IC} = 0.044$ as a PASS based on the actual SE. The daily-IC framework resolves this: the empirical $\mathrm{SE} = 0.00560$ (not 0.0366 from Fisher $z$) gives $\mathrm{IC}_{\min} = 0.0157$, which aligns naturally with the 0.02 threshold.

### 2.2 Power Analysis

For an observed IC of $r$, the $t$-statistic is

$$t(r) = \frac{|r|}{\mathrm{SE}(\mu_{\mathrm{IC}})} = \frac{|r|}{0.00560}. \tag{10}$$

Table 2: Power to detect various IC values at $\alpha = 0.05$ (two-sided). $\mathrm{SE} = 0.00560$, $T = 795$.

| IC $r$ | $t = r/\,\mathrm{SE}$ | $p$-value (two-sided) | Power |
|---|---|---|---|
| 0.005 | 0.89 | 0.37 | 0.14 |
| 0.010 | 1.79 | 0.07 | 0.45 |
| 0.016 | 2.80 | 0.005 | 0.80 |
| 0.020 | 3.57 | $3.6 \times 10^{-4}$ | 0.94 |
| 0.029 | 5.15 | $2.7 \times 10^{-7}$ | >0.999 |
| 0.044 | 7.86 | $4.0 \times 10^{-15}$ | >0.999 |

Thus the threshold $|\,\mathrm{IC}\,| < 0.02$ identifies the regime where the signal is *barely above the detection floor*: at $\mathrm{IC} = 0.02$, power is 94% (adequate), but at $\mathrm{IC} = 0.01$, power drops to 45% (inadequate). The threshold provides a conservative margin over the minimum detectable IC of 0.016.

**Proposition 2.2** (Threshold 1: Detection)**.**

$$\boxed{|\,\mathrm{IC}\,| < 0.02 \implies \text{Below detection}} \tag{11}$$

This is a principled threshold: it is $1.28 \times \mathrm{IC}_{\min}$, providing a 28% safety margin while staying aligned with the empirical SE.

## 3 Threshold 2: Held-Out IC Gap

### 3.1 Framework

Let $\mu_{\mathrm{IC}}^{(\mathrm{train})}$ and $\mu_{\mathrm{IC}}^{(\mathrm{held})}$ be the mean daily ICs on the training and held-out sets, respectively, and

$$\Delta = \mu_{\mathrm{IC}}^{(\mathrm{train})} - \mu_{\mathrm{IC}}^{(\mathrm{held})}. \tag{12}$$

Under $H_0$ (no overfitting), both estimate the same population quantity, so $\mathbb{E}[\Delta] = 0$.

With the daily-IC framework, the variance of each mean is

$$\mathrm{Var}\big(\mu_{\mathrm{IC}}^{(k)}\big) = \frac{(\sigma_{\mathrm{IC}}^{(k)})^2}{T_k}, \qquad k \in \{\mathrm{train}, \mathrm{held}\}. \tag{13}$$

## 3.2 Bounding the Standard Error of the Gap

Since training and held-out sets are disjoint in time, their daily IC sequences are serially separated by the training/held-out boundary. Under the assumption of weak temporal dependence (autocorrelation decays within a few lags), the covariance term is negligible:

$$\mathrm{SE}(\Delta) = \sqrt{\frac{(\sigma_{\mathrm{IC}}^{(\mathrm{train})})^2}{T_{\mathrm{train}}} + \frac{(\sigma_{\mathrm{IC}}^{(\mathrm{held})})^2}{T_{\mathrm{held}}} - 2\ \mathrm{Cov}(\mu_{\mathrm{IC}}^{(\mathrm{train})},\ \mu_{\mathrm{IC}}^{(\mathrm{held})})}. \tag{14}$$

A *conservative* estimate uses the same $\sigma_{\mathrm{IC}}$ for both sets and assumes zero covariance:

$$\mathrm{SE}(\Delta) \approx \sigma_{\mathrm{IC}} \sqrt{\frac{1}{T_{\mathrm{train}}} + \frac{1}{T_{\mathrm{held}}}}. \tag{15}$$

With the empirical values ($\sigma_{\mathrm{IC}} = 0.158$, $T_{\mathrm{train}} = 1853$, $T_{\mathrm{held}} = 795$):

$$\begin{aligned}
\mathrm{SE}_{\mathrm{train}} &= \frac{0.158}{\sqrt{1853}} = 0.00367, \\
\mathrm{SE}_{\mathrm{held}} &= \frac{0.158}{\sqrt{795}} = 0.00560, \\
\mathrm{SE}(\Delta) &= \sqrt{0.00367^2 + 0.00560^2} = \sqrt{1.347\times10^{-5} + 3.136\times10^{-5}} \\
&= \sqrt{4.483\times10^{-5}} = \mathbf{0.00670}.
\end{aligned} \tag{16}$$

Table 3: Gap diagnostics in the daily-IC framework.

| Gap $\Delta$ | $\Delta/\,\mathrm{SE}(\Delta)$ | One-sided $p$ | Classification |
|---|---|---|---|
| 0.005 | 0.75 | 0.23 | Within noise |
| 0.010 | 1.49 | 0.068 | Marginally suspicious |
| 0.020 | 2.99 | 0.0014 | Significant overfitting |
| 0.050 | 7.46 | $< 10^{-13}$ | Pathological |

**Proposition 3.1** (Threshold 2: Gap categories)**.**

$$\boxed{\Delta < 0.005} \quad \text{(Healthy — within 1 SE)}, \tag{17}$$

$$\boxed{0.005 \le \Delta \le 0.020} \quad \text{(Suspicious — 1–3 SE)}, \tag{18}$$

$$\boxed{\Delta > 0.020} \quad \text{(Pathological — exceeds 3 SE)}. \tag{19}$$

These thresholds are tighter than the Revision 1 values (0.01, 0.05) because the daily-IC SE is substantially smaller than the Fisher-$z$ SE (which assumed $N = 750$ effective observations). The Revision 1 thresholds corresponded to $0.24$–$1.18\sigma$; the new thresholds are properly calibrated to $0.75$–$2.99\sigma$.

## 3.3 Per-Window Gap

For the walk-forward per-window gap used in Step 5, the validation window contains $T_{\mathrm{val}} \approx 252$ days:

$$\mathrm{SE}(\Delta^{(w)}) \approx 0.158 \sqrt{\frac{1}{1008} + \frac{1}{252}} = 0.158 \times 0.0704 = 0.0111. \tag{20}$$

The threshold $\bar{\Delta}^{(w)} < 0.03$ (used in code) corresponds to $0.03/0.0111 = 2.70$ SE, a conservative bound.

## 4 Threshold 3: Coefficient of Variation of IC Across Windows

### 4.1 Framework

For $W$ walk-forward windows, let the window-$w$ IC be the mean of daily ICs within that window's validation period:

$$\mathrm{IC}_w = \frac{1}{T_w} \sum_{t \in \mathcal{T}_w} \hat{\rho}_t, \qquad w = 1, \dots, W. \tag{21}$$

The CV is

$$\mathrm{CV} = \frac{s_{\mathrm{IC}}}{|\bar{\mathrm{IC}}|}, \qquad \bar{\mathrm{IC}} = \frac{1}{W} \sum_{w=1}^{W} \mathrm{IC}_w, \qquad s_{\mathrm{IC}}^2 = \frac{1}{W-1} \sum_{w=1}^{W} (\mathrm{IC}_w - \bar{\mathrm{IC}})^2. \tag{22}$$

### 4.2 Expected CV Under $H_0 : \mu_{\mathrm{IC}} = 0$

This is where Revision 1 contained a critical algebraic error. Using $\sigma_0^2 \approx 1/N$ with $N \approx 10^5$ (pooled pairs) produced $\sigma_0 \approx 0.003$, leading to the incorrect conclusion that $\mathbb{E}[\mathrm{CV} \mid H_0] \gg 1$ was *obvious*. In reality, the window-IC standard error under $H_0$ is determined by the daily-IC framework, not by the pooled Fisher-$z$ variance.

**Proposition 4.1** (Expected CV under the null)**.** Under $H_0$, each window IC is a mean of $T_w$ approximately independent daily ICs:

$$\mathrm{IC}_w \sim \mathcal{N}\left(0,\ \frac{\sigma_{\mathrm{IC}}^2}{T_w}\right). \tag{23}$$

For $W$ windows of equal size $T_w = T_{\mathrm{val}}$,

$$\bar{\mathrm{IC}} \sim \mathcal{N}\left(0,\ \frac{\sigma_{\mathrm{IC}}^2}{T_{\mathrm{val}} W}\right), \qquad s_{\mathrm{IC}} \approx \frac{\sigma_{\mathrm{IC}}}{\sqrt{T_{\mathrm{val}}}}. \tag{24}$$

Since $|\bar{\mathrm{IC}}|$ is $O_p(1/\sqrt{W})$ while $s_{\mathrm{IC}}$ is $O_p(1)$ (relative to $\sigma_{\mathrm{IC}}/\sqrt{T_{\mathrm{val}}}$),

$$\mathbb{E}[\mathrm{CV} \mid H_0] \approx \frac{\sigma_{\mathrm{IC}}/\sqrt{T_{\mathrm{val}}}}{\sigma_{\mathrm{IC}}/\sqrt{T_{\mathrm{val}} W} \cdot \sqrt{2/\pi}} = \sqrt{\frac{\pi W}{2}} \approx 4.16 \quad (\text{for } W = 11). \tag{25}$$

Thus, under the null hypothesis of no predictive signal, we *expect* $\mathrm{CV} \approx 4$–$5$, not $\mathrm{CV} \gg 1$ as Revision 1 implied. This means:

- $\mathrm{CV} < 1$ is already *strong evidence* against $H_0$.
- $\mathrm{CV} = 0.6$ is *conclusive* evidence of non-zero IC.
- The Revision 1 CV thresholds ($< 0.3$, $0.3$–$0.5$, $> 0.6$) are actually *correct* despite the flawed derivation; the daily-IC framework provides the missing rigorous justification.

### 4.3 Boundary Signal Detection at $\mathrm{CV} = 0.6$

**Proposition 4.2** (CV as inverse SNR)**.** For a genuine signal with $\mu_{\mathrm{IC}} > 0$ and window-level variation $\sigma_{\mathrm{w}}$,

$$\mathrm{CV} = \frac{\sigma_{\mathrm{w}}}{\mu_{\mathrm{IC}}}, \qquad \mathrm{SNR}_{\mathrm{window}} = \frac{1}{\mathrm{CV}}. \tag{26}$$

At $\mathrm{CV} = 0.6$, $\mathrm{SNR} = 1.667$.

- $\mathbb{P}(\mathrm{IC}_w > 0) = \Phi(\mathrm{SNR}) = \Phi(1.667) = 0.952$.
- $t$-statistic for $\bar{\mathrm{IC}}$: $t = \bar{\mathrm{IC}}/(s_{\mathrm{IC}}/\sqrt{W}) = \sqrt{W}/\,\mathrm{CV} = \sqrt{11}/0.6 = 5.53$, $p < 10^{-3}$ (one-sample).

Table 4: CV thresholds and implications.

| CV | SNR | $\mathbb{P}(\text{IC}_w > 0)$ | $t_{W=11}$ | Classification |
|---|---|---|---|---|
| $< 0.3$ | $> 3.33$ | $> 0.9996$ | $> 11.1$ | Healthy |
| 0.3–0.5 | 2.0–3.33 | 0.977–0.9996 | 6.6–11.1 | Suspicious |
| 0.5–0.6 | 1.67–2.0 | 0.952–0.977 | 5.5–6.6 | Marginal pass |
| $> 0.6$ | $< 1.67$ | $< 0.952$ | $< 5.5$ | Failing |
| $> 1.0$ | $< 1.0$ | $< 0.841$ | $< 3.3$ | Noise-dominated |

# 5 Thresholds 4 & 6: Daily IC Positive Fraction (Replaces Per-Stock Binomial)

## 5.1 The Problem with Per-Stock Binomial Tests

Revision 1's per-stock binomial test assumes independent stock-level ICs. This assumption is *known to be violated* in financial data: factor exposures and sector co-movements induce positive cross-sectional correlation, which inflates the variance of $N_+$ and makes the binomial test *anti-conservative* (rejects $H_0$ too easily).

The daily-IC framework offers a cleaner, self-consistent alternative.

## 5.2 Daily IC Positive Fraction

**Definition 5.1** (Daily IC positive fraction)**.**

$$p_+ = \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}\{\hat{\rho}_t > 0\}. \tag{27}$$

This is the fraction of trading days on which the cross-sectional Spearman IC is positive.

Under $H_0$ (no predictive ability), daily ICs are symmetrically distributed around zero, so $p_+ = 0.5$ in expectation. Under a genuine positive signal, $p_+ > 0.5$.

## 5.3 Binomial Test on Daily Indicators

If daily IC signs are approximately independent,[1] then

$$T_+ = \sum_{t=1}^{T} \mathbf{1}\{\hat{\rho}_t > 0\} \sim \text{Binomial}(T,\, p). \tag{28}$$

Under $H_0 : p = 0.5$:

$$z = \frac{\hat{p}_+ - 0.5}{\sqrt{0.25/T}} = \frac{\hat{p}_+ - 0.5}{\sqrt{0.25/795}} = \frac{\hat{p}_+ - 0.5}{0.01773}. \tag{29}$$

## 5.4 Handling Autocorrelation in IC Signs

If daily IC signs exhibit serial dependence (e.g., consecutive positive days cluster during bull markets), the effective sample size for the binomial test is reduced. Using a runs-test-based correction:

$$T_{\text{eff}} = T \times \frac{2\, R_+ R_-}{R_+ + R_-} \times \frac{1}{T}, \tag{30}$$

[1] Serial dependence in daily IC signs is typically much weaker than dependence in magnitudes, making the binomial approximation more defensible than for per-stock tests. The Newey–West adjustment for sign-autocorrelation is discussed below.

Table 5: Statistical significance of daily IC positive fractions ($T = 795$).

| $\hat{p}_+$ | $T_+$ | $z$ | $p$ (one-sided) |
|---|---|---|---|
| 0.500 | 398 | 0.00 | 0.500 |
| 0.520 | 413 | 1.13 | 0.130 |
| 0.550 | 437 | 2.82 | 0.0024 |
| 0.570 | 453 | 3.95 | $3.9 \times 10^{-5}$ |
| 0.600 | 477 | 5.64 | $8.5 \times 10^{-9}$ |
| 0.650 | 517 | 8.46 | $< 10^{-16}$ |

where $R_+$ and $R_-$ are the numbers of runs of positive and negative signs, respectively. In the worst case of complete clustering (all positives together), $T_{\text{eff}} \approx 2$, rendering the test uninformative. In practice, financial IC sign sequences are well-mixed (runs-test $p > 0.1$ in most pipelines), so $T_{\text{eff}} \approx T$.

A more general approach uses Newey–West on the sign indicators themselves:

$$\mathrm{SE}_{\mathrm{NW}}(\hat{p}_+) = \frac{1}{T}\sqrt{\sum_{t=1}^{T}(Y_t - \hat{p}_+)^2 + 2\sum_{k=1}^{L} w_k \sum_{t=k+1}^{T}(Y_t - \hat{p}_+)(Y_{t-k} - \hat{p}_+)}, \tag{31}$$

where $Y_t = \mathbf{1}\{\hat{\rho}_t > 0\}$.

### 5.5 Proposed Thresholds

**Proposition 5.1** (Threshold 4: Daily IC positive fraction)**.**

$$\boxed{\hat{p}_+ > 0.55} \quad \text{(Healthy — } z > 2.82,\ p < 0.0025\text{)}, \tag{32}$$

$$\boxed{0.50 \le \hat{p}_+ \le 0.55} \quad \text{(Suspicious — marginal evidence)}, \tag{33}$$

$$\boxed{\hat{p}_+ < 0.50} \quad \text{(Pathological — worse than chance)}. \tag{34}$$

**Proposition 5.2** (Threshold 6: Binary pass/fail)**.** The code's binary pass criterion becomes:

$$\boxed{\text{pass\_daily\_positive} = (\hat{p}_+ > 0.55)}. \tag{35}$$

For the pipeline v2 Ridge with $\hat{p}_+ = 0.570$, this is a comfortable PASS ($z = 3.95$, $p = 3.9 \times 10^{-5}$).

### 5.6 Comparison with Per-Stock Binomial

The daily-IC positive-fraction test is superior on three grounds:

1. **Consistent unit**: same daily cross-section used across all thresholds.
2. **Reduced dependence**: daily IC signs are less cross-sectionally correlated than per-stock ICs (which share market-wide factor exposures across all windows).
3. **Higher statistical power**: $T = 795$ daily observations vs. $S = 200$–$477$ stocks, with the added benefit that each daily IC aggregates information from all $N$ stocks.

## 6 Threshold 5: CV $< 0.6$ as Pass

### 6.1 CV in the Daily-IC Framework

The code's CV pass criterion is $\mathrm{CV}(\mathrm{IC}) < 0.6$, where IC is the vector of $W$ walk-forward window ICs. As shown in §4, this corresponds to $\mathrm{SNR} > 1.67$ and $\mathbb{P}(\mathrm{IC}_w > 0) > 0.952$.

### 6.2 Illustrative Computation with Pipeline v2 Ridge

Pipeline v2 Ridge yields $W = 11$ window ICs with $\bar{\mathrm{IC}} = 0.0648$, $s_{\mathrm{IC}} = 0.0382$, and $\mathrm{CV} = 0.590$:

$$\begin{aligned}
\mathrm{SNR}_{\mathrm{window}} &= 1/0.590 = 1.695, & (36)\\
\mathbb{P}(\mathrm{IC}_w > 0) &= \Phi(1.695) = 0.955, & (37)\\
t\text{-stat} &= \sqrt{11}/0.590 = 5.62, & (38)\\
p\text{-value} &= 2.2 \times 10^{-4}. & (39)
\end{aligned}$$

### 6.3 Connection to Kelly–Malamud LLG Bound

The Kelly–Malamud [3] LLG (Leverage, Loss, Grid) bound provides a fundamental upper limit on the ratio of predictable to total return variance, $P/T$. In the daily-IC framework:

$$\frac{P}{T} = \frac{\sigma^2_{\mathrm{pred}}}{\sigma^2_{\mathrm{total}}}. \tag{40}$$

Decompose the observed window-IC variance into signal and noise components:

$$\mathrm{Var}(\mathrm{IC}_w) = \underbrace{\mathrm{Var}(\mu^{(w)}_{\mathrm{IC}})}_{\text{signal}} + \underbrace{\frac{\sigma^2_{\mathrm{IC}}}{T_{\mathrm{val}}}}_{\text{estimation noise}}, \tag{41}$$

$$\sigma^2_{\mathrm{sig}} = s^2_{\mathrm{IC}} - \frac{\sigma^2_{\mathrm{IC}}}{T_{\mathrm{val}}}. \tag{42}$$

For pipeline v2 Ridge ($s_{\mathrm{IC}} = 0.0382$, $\sigma_{\mathrm{IC}} = 0.158$, $T_{\mathrm{val}} = 252$):

$$\begin{aligned}
\frac{\sigma^2_{\mathrm{IC}}}{T_{\mathrm{val}}} &= \frac{0.158^2}{252} = \frac{0.02496}{252} = 9.91 \times 10^{-5}, & (43)\\
\sigma^2_{\mathrm{sig}} &= 0.0382^2 - 9.91 \times 10^{-5} = 0.001459 - 0.000099 = 0.001360, & (44)\\
\frac{P}{T} &= \frac{0.001360}{0.001459} = 0.932. & (45)
\end{aligned}$$

Thus over 93% of window-IC variance is attributable to genuine cross-window signal variation, not estimation noise. This aligns with the CV-based conclusion that the pipeline produces consistent, non-spurious predictions.

## 7 Multiple Testing Correction

The diagnostic framework applies up to 11 pass/fail checks across 5 steps. Without correction, the family-wise error rate (FWER) at $\alpha = 0.05$ is inflated.

### 7.1 Bonferroni Correction

**Proposition 7.1** (Bonferroni-adjusted significance level)**.** For $m = 11$ tests and a target FWER of $\alpha_{\mathrm{FW}} = 0.05$,

$$\boxed{\alpha' = \frac{\alpha_{\mathrm{FW}}}{m} = \frac{0.05}{11} = 0.00455}. \tag{46}$$

All individual-test $p$-values should be compared to $\alpha'$, not 0.05 or 0.01.

**Impact on thresholds:**

- **Step 2 $t$-test**: $p < 0.00455$ (previously $p < 0.01$).

- **Step 3 binomial**: $p < 0.00455$ (previously $p < 0.01$). With $W = 11$, this requires 10/11 or 11/11 positive windows.
- **Step 4 binomial**: $p < 0.00455$. For $T = 795$, this requires $\hat{p}_+ > 0.548$ (previously 0.55, essentially unchanged).
- **Step 4 $t$-test**: $p < 0.00455$ (previously $p < 0.01$).

The daily-IC-based Threshold 4 ($\hat{p}_+ > 0.55$) is nearly unchanged by the correction, while the per-window binomial test (requiring $p < 0.01$ on $W = 11$) becomes stricter.

### 7.2 Hierarchical Gatekeeping

A more powerful alternative to Bonferroni uses hierarchical testing:

1. **Steps 1–2 are gatekeepers**: the model must pass the held-out IC detection test *and* the $t$-test of $\mu_{\text{IC}} > 0$ at $\alpha = 0.05$.
2. **Steps 3–5 are applied only if gatekeepers pass**. Within Steps 3–5, apply Bonferroni correction for the remaining 9 tests ($\alpha' = 0.05/9 = 0.00556$).
3. **Overall verdict**: PASS requires all gatekeepers passed *and* at least 7/9 secondary tests passed (after correction).

This hierarchical structure reflects the logical dependence: if the model has no detectable signal (Steps 1–2), stability and overfitting checks are moot.

## 8 Implementation Checklist

### 8.1 Computing the Daily IC Time-Series

The daily IC array is already computed in `diagnose_v2.py` (lines 190–209) as `ho_daily_ic`. The existing code already computes `daily_ic_mean`, `daily_ic_std`, and `daily_ic_positive_frac` and stores them in the JSON output.

**Extension needed:** compute daily ICs for each walk-forward window's validation period, to enable per-window daily-IC analysis.

### 8.2 Computing Newey–West Standard Errors

```
import numpy as np
from scipy import stats

def newey_west_se(ts, max_lag=None):
    """HAC standard error of the mean (Bartlett kernel)."""
    T = len(ts)
    if max_lag is None:
        max_lag = int(4 * (T / 100) ** (2 / 9))
    mu = np.mean(ts)
    gamma = np.zeros(max_lag + 1)
    for k in range(max_lag + 1):
        gamma[k] = np.mean(
            (ts[k:] - mu) * (ts[:T-k] - mu)
        )
    w = 1 - np.arange(1, max_lag + 1) / (max_lag + 1)
    long_run_var = gamma[0] + 2 * np.sum(w * gamma[1:])
    return np.sqrt(long_run_var / T)
```

### 8.3 Updating the Verdict Thresholds

The hard-coded thresholds in `diagnose_v2.py` should be replaced with data-driven equivalents computed from the daily IC time-series:

Table 6: Threshold migration from Revision 1 to Revision 2.

| Check | Rev. 1 Value | Rev. 2 Value | Derivation |
|---|---|---|---|
| Step 1 pass | $\lvert \mathrm{IC} \rvert > 0.02$ | $\lvert \mathrm{IC} \rvert > 0.02$ | §2, Eq. (9) |
| Step 2 pass | $p < 0.01$ | $p < 0.00455$ | §7, Eq. (46) |
| Step 3 CV pass | $\mathrm{CV} < 0.6$ | $\mathrm{CV} < 0.6$ | §4, unchanged but re-derived |
| Step 3 binomial pass | $p < 0.01$ | $p < 0.00455$ | Bonferroni correction |
| Step 3 recency pass | $\Delta > -0.03$ | $\Delta > -0.02$ | §3, $\approx 2$ SE($\Delta^{(w)}$) |
| Step 4 binomial pass | $p < 0.01$ | $\hat{p}_+ > 0.55$ | §5, replaces per-stock |
| Step 4 $t$-test | $p < 0.01$ | $p < 0.00455$ | Bonferroni correction |
| Step 4 majority | $\hat{p} > 0.55$ | $\hat{p}_+ > 0.55$ | §5, daily framework |
| Step 5 gap pass | $\bar{\Delta}^{(w)} < 0.03$ | $\bar{\Delta}^{(w)} < 0.02$ | §3, $\approx 1.8$ SE |
| Step 5 decay pass | $\Delta_{\text{half}} > -0.02$ | $\Delta_{\text{half}} > -0.01$ | Tighter, daily-IC calibrated |
| Step 5 bootstrap pass | $\mathrm{CI}_{\text{lower}} > 0.02$ | $\mathrm{CI}_{\text{lower}} > 0.016$ | §2, $\mathrm{IC}_{\text{min}}$ |

### 8.4 Validation Protocol

After implementing Revision 2 thresholds, verify against the pipeline v2 Ridge result (IC $= 0.065$, CV $= 0.59$, $\hat{p}_+ = 0.57$):

- The Ridge pipeline should remain a clear PASS (as expected for a well-established baseline).
- Apply to StockCTM (IC $= 0.056$–$0.062$, borderline) and MambaStock (IC $\approx 0$, expected FAIL) as sanity checks.
- Compare verdicts between Revision 1 and Revision 2 to ensure no unexpected threshold reversals for documented results.

## 9 Summary: Comparison with Revision 1

## References

## References

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Table 7: Framework comparison: Revision 1 vs. Revision 2.

| Aspect | Revision 1 | Revision 2 (this document) |
|---|---|---|
| **Unit of analysis** | Pooled Spearman $\rho$ over all stock–day pairs | Daily cross-sectional Spearman $\rho_t$ time-series |
| SE(IC) | Fisher $z$: $1/\sqrt{N-3}$ with inconsistent $N$ (750; 100K; 375K) | $\sigma_{\mathrm{IC}}/\sqrt{T}$ with consistent $\sigma_{\mathrm{IC}} = 0.158$, $T = 795$ |
| **Threshold 1 derivation** | Fisher $z \to r_{\min} = 0.102$, then *ad hoc* threshold 0.02 | Power analysis $\to \mathrm{IC}_{\min} = 0.0157$, principled threshold 0.02 |
| **Threshold 2 gap** | Two opposite SE estimates (0.042 and 0.002) used in same section | Single consistent SE = 0.0067, from daily-IC $\sigma_{\mathrm{IC}}$ |
| **Threshold 3 CV** | $\sigma_0^2 \approx 10^{-5}$ (pooled pairs) $\to \mathrm{CV}_{\mathrm{null}} \gg 1$ | $\sigma_0^2 = \sigma_{\mathrm{IC}}^2/T_w$ (daily ICs) $\to \mathrm{CV}_{\mathrm{null}} \approx 4.2$ |
| **Per-stock binomial** | $S = 200$ independent stocks; ignores cross-sectional correlation | Replaced by daily-IC positive fraction $\hat{p}_+$ with $T = 795$ daily observations |
| **Multiple testing** | Not addressed | Bonferroni $\alpha' = 0.05/11 = 0.00455$; hierarchical gatekeeping alternative |
| **Theoretical foundation** | Fisher $z$-transformation (bivariate normality assumption) | Central Limit Theorem for daily-IC means (valid under weak dependence) |

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